DAVID BEINHAUER

GitHub: dbeinhauer ⋄ LinkedIn
Phone: (+420) 777-186-273 ⋄ Email: david.beinhauer@email.cz

RESEARCH INTERESTS

Aspiring PhD candidate with an interdisciplinary background in computer science and bioinformatics, aiming to advance the field of computational neuroscience. I am particularly interested in applying machine learning techniques to model neural systems, with a focus on developing technologies for neural restoration through brain-machine interfaces.

EDUCATION

M.Sc. in Bioinformatics, Charles University, Prague, Czech Republic — 2022–2025
GPA: **1.33** (1–4 | 1 is best)
Relevant Coursework: Computational Neuroscience, Spatio-Temporal Modeling of Biological Systems, Cell Biology

B.Sc. in Computer Science, Charles University, Prague, Czech Republic — 2019–2022
Specialization: **Artificial Intelligence**
GPA: **1.28** (scale: 1–4 | 1 is best)
Relevant Coursework: Machine Learning, Deep Learning, Statistics, Data Analysis

RESEARCH EXPERIENCE

Research Assistant, Czech Academy of Sciences,
National Institute of Mental Health — Dec 2025 – Present

- Conducting analysis and comparative evaluation of EEG data from participants receiving psilocybin versus placebo during music listening sessions.
- Investigating inter-subject neural synchrony in response to auditory stimulation.
- Examining distinctive patterns of neural responses to music under the influence of psilocybin.

Modeling Visual Cortex with Biologically Inspired Recurrent Neural Networks

- **Master Thesis:** GitHub (Code) / GitHub (Thesis)
- Developed a biologically realistic model of the primary visual cortex (V1) using modular, multi-layered recurrent neural networks.
- Mapped synthetic neurons on a one-to-one basis with those from a state-of-the-art spiking neural network model of V1.
- Demonstrated that incorporating biological realism is essential for capturing the temporal dynamics of the visual system.
- Mentored a fellow bachelor student on extending the core research.

TALKS AND CONFERENCE PRESENTATIONS

[1] **Beinhauer, D.**, Kraus R, Baroni L, Antolík J (2025). *Bridging Biology and Deep Learning: Modeling Visual Cortex with Structured Recurrent Networks.* Bernstein Conference 2025. doi: 10.12751/nncn.bc2025.151

[2] **Beinhauer, D.** (2025, Nov 12) *Bridging Biology and Deep Learning: Modeling Visual Cortex with Structured Recurrent Networks.* Oral research pitch. Young AI Research Forum 2025: AI as Driver for Interdisciplinary Research, Prague, Czech Republic.

[3] **Beinhauer, D.**, Kraus, R., Baroni, L., Antolík, J. (2025, May 27). *Advancing visual neuroscience with biologically inspired recurrent neural networks.* [Poster presentation by J. Antolík]. The Sixth International Conference on the Mathematics of Neuroscience and AI. Split, Croatia.

INDUSTRY RESEARCH EXPERIENCE

Bioinformatics Internship, MSD Czech Republic Dec 2025 – Present

- Member of the Bioinformatics team, focusing on developing computational solutions enhancing drug discovery and development processes.
- Applied state-of-the-art and machine learning techniques to enhance protein structural predictions.

NLP Internship, MSD Czech Republic Oct 2023 – Nov 2025

- Member of the Artificial Intelligence team, focused on Natural Language Processing (NLP) applications in the pharmaceutical domain.
- Designed and implemented Large Language Model (LLM)-based solutions for scientific document processing and data retrieval.
- Collaborated with medical writers to integrate LLM techniques into the research and drafting of internal medical documentation.
- Contributed to the development of a modular NLP framework using state-of-the-art technologies and best software engineering practices.
- Regularly communicated technical solutions to both expert and non-expert stakeholders to promote adoption of cutting-edge NLP methods.

AWARDS AND HONORS

Travel Grant to the Bernstein Conference 2025

- Awarded a travel grant as an early-career scientist presenting as first author in the poster session at the Bernstein Conference 2025.

Nomination for Dean's Award 2025

- Nominated for one of the best final theses of the academic year at the faculty, an award recognizing outstanding research quality and contribution.

Scholarship for Academic Excellence 2021, 2023

- Awarded twice for ranking among the top 10% of students based on academic performance during both bachelor's and master's studies.

SKILLS

Technical Skills

- Programming Languages: Python, C++
- Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-learn
- Cloud and Computing Tools: AWS, HPC environments, Docker
- Development Tools: Git, Jira

Theoretical and Analytical Skills

- Computer Science: Machine Learning (Deep Learning, Natural Language Processing), Statistical Analysis, Data Processing for Biological Systems
- Computational Neuroscience: Visual System Modeling, EEG Signal Analysis
- Biological Sciences: Cell Biology, Genomics, Structural Biology, Epigenetics, Basic Neurobiology

Languages

- English: Professional proficiency
- Czech: Native speaker